

Relational-Indexed Retrieval-Augmented Generation (RI-RAG):

Scalable Retrieval Efficiency and Memory Consumption Evaluation

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1 Research Status and Problem to Be Solved

Relational-Indexed Retrieval-Augmented Generation (RI-RAG) has already been developed as a working minimum viable prototype. Its architecture separates semantic retrieval from authoritative data storage: a vector database is used to identify relevant records, while the corresponding textual payload and relational metadata are stored in a relational database.

An initial experimental phase has also been completed. Using a 13-record controlled corpus, the prototype demonstrated measurable reductions in the materialized vector-store representation and in process-level memory consumption compared with a conventional RAG configuration.

The refined benchmark measured:

- a **24.70% reduction** in the measured materialized Chroma representation;
- a **31.01% lower Python-process peak RSS increase**;
- a **12.65% lower combined peak RSS increase** when the evaluated Python and MySQL processes were considered together; and
- **90% top-1 retrieval accuracy** on a 10-query ground-truth evaluation.

These results provide preliminary evidence for the architectural hypothesis, but the current corpus is too small to establish scalability. The next research phase is therefore focused on determining whether the observed behavior persists as the number of records and payload sizes increase.

The next phase will investigate two primary questions:

1. **Retrieval efficiency:** Can RI-RAG retrieve relevant information within a competitive time compared with conventional vector-database RAG as the dataset grows?
2. **System-level memory consumption:** Does reducing the amount of payload stored in the vector database translate into a measurable reduction in the overall RAM footprint of the complete RAG system, rather than only reducing the memory associated with the vector-store representation?

The research will additionally evaluate storage requirements, retrieval accuracy, semantic-summary quality, database hydration overhead, and scalability.

The immediate large-scale experimental phase is currently **paused pending approximately \$300 in funding for AI/LLM usage and dataset-generation assistance**. This funding is intended to support the generation, refinement, validation, and processing of the substantially larger experimental datasets required for the next stage.

The existing RI-RAG implementation, benchmark infrastructure, MVP, and preliminary results are already available. The requested funding is therefore intended to enable the next experimental scale rather than to establish the project itself.

2 Completed Preliminary Research

The first research phase established the RI-RAG prototype and evaluated it against a conventional RAG configuration using a controlled 13-record corpus.

Table 1: Completed preliminary RI-RAG results.

Measurement	Result
Materialized vector-store representation reduction	24.70%
Python-process peak RSS increase reduction	31.01%
Combined Python + MySQL peak RSS increase reduction	12.65%
Top-1 retrieval accuracy	90.0%
Ground-truth queries	10
Corpus size	13 records

The 12.65% combined result is the principal process-level memory measurement because it accounts for the additional MySQL process required by RI-RAG. The result should not be interpreted as a universal reduction in total infrastructure memory.

The preliminary evaluation also identified a retrieval-latency trade-off. The current isolated-process benchmark measured mean retrieval latency of 135.81 ms for conventional RAG and 216.32 ms for RI-RAG. Because these measurements were obtained from limited trials and exhibited substantial variation, they are considered preliminary and require repeated controlled experiments.

The initial study therefore established a working experimental baseline while identifying the principal questions requiring larger-scale investigation.

3 Research Objectives

The next research phase has five principal objectives:

1. Determine whether RI-RAG’s measured memory reduction persists at 10,000 and 100,000 records.
2. Measure whether vector-store memory reduction translates into lower overall system-level RAM consumption.

3. Characterize the retrieval-latency cost of relational record hydration at increasing dataset sizes.
4. Validate the semantic relationship between generated `content` and `semantic.summary` fields.
5. Quantify the trade-offs among memory, storage, latency, retrieval accuracy, and scalability.

The research will not assume that RI-RAG will outperform conventional RAG on every metric. The purpose of the next phase is to measure the conditions under which the architecture provides resource benefits and to quantify any associated costs.

4 Expected Resource Usage

4.1 Datasets

Two principal dataset scales are planned:

- **10,000 records:** initial large-scale validation and benchmarking.
- **100,000 records:** scalability evaluation at a substantially larger dataset size.

The datasets will be generated and refined using AI assistance. This is necessary because the RI-RAG experiment uses a strict database projection that must be consistently populated across a large number of records. Manually creating and validating tens of thousands of records would require substantial time and could introduce inconsistencies in the experimental data.

Each record will follow the RI-RAG relational projection:

Table 2: RI-RAG relational projection.

Field	Description
<code>file_id</code>	Unique identifier for the record.
<code>file_name</code>	Name associated with the record.
<code>content</code>	Full textual content stored in the relational database.
<code>semantic.summary</code>	Concise semantic representation of the content.
<code>grade</code>	Evaluation or quality score associated with the record.
<code>comments</code>	Additional evaluation comments associated with the record.

The AI-assisted dataset-generation pipeline will focus on generating records that conform to this schema while maintaining consistency between textual content, semantic summaries, grades, comments, and record identifiers.

4.2 Content–Summary Validation

A specific validation experiment will measure the semantic relationship between each record’s `content` and its `semantic.summary`.

Both fields will be converted into embeddings using the same embedding model. Their cosine similarity will then be calculated to estimate how closely the generated summary represents the original content.

For each dataset, the experiment will report:

- Mean cosine similarity.
- Median similarity.
- Standard deviation.
- Similarity distribution.
- Lower and upper percentile values.
- Percentage of records meeting the selected validation criterion.

For a record i :

$$S_i = \frac{E(c_i) \cdot E(s_i)}{\|E(c_i)\| \|E(s_i)\|}$$

where $E(c_i)$ represents the embedding of the original content and $E(s_i)$ represents the embedding of its semantic summary.

Semantic similarity will be treated as a validation signal rather than proof of factual correctness. Where appropriate, a smaller sample may also undergo additional AI-assisted content-coverage or factual-consistency evaluation.

The generated data will also be validated for:

- Correct database schema compliance.
- Valid and unique `file_id` values.
- Presence of all required fields.
- Consistency between `content` and `semantic_summary`.
- Valid `grade` values.
- Appropriate `comments`.
- Successful ingestion into both relational and vector components.

4.3 AI-Assisted Research and Immediate Funding Requirement

The project will use AI assistance primarily for dataset generation and refinement, coding assistance, database optimization, and experimental preparation.

An immediate allocation of approximately **\$300** is requested for AI and LLM usage.

This funding will support:

- AI-assisted generation of the 10,000- and 100,000-record datasets.

- Dataset refinement and consistency checking.
- Coding assistance during optimization.
- Database and retrieval optimization.
- Content–summary semantic validation.
- Experimental preparation and analysis.

The exact token usage will depend on the generation and validation strategy and will be measured during the project.

The large-scale research phase is currently paused until these resources become available.

4.4 Human Research Assistance

The project is currently developed independently. At the optimization stage, an additional researcher or technical assistant could help accelerate dataset preparation, testing, optimization, and validation.

Human research assistance is therefore identified as a potential future resource rather than an immediate prerequisite for restarting the experiment.

If appropriate, part of future research funding may support assistance during the optimization and validation phase.

4.5 Researcher Stipend

A **12-month research stipend of \$500 per month** is proposed as a separate funding requirement, for a total of **\$1,500**.

The stipend would support continued full-time work on RI-RAG, including:

- Development and optimization of the implementation.
- Dataset preparation and experimental validation.
- Benchmarking against conventional RAG.
- Memory and latency analysis.
- Investigation of payload-size effects.
- Analysis of memory/latency trade-offs.
- Documentation and dissemination.

The \$1,500 stipend represents the proposed resource requirement for the broader 3-month research program and is distinct from the immediate \$300 required to resume large-scale experimentation.

Table 3: Proposed project budget.

Category	Allocation
AI assistance and LLM usage	\$300
3-month research stipend	\$1,500
Total	\$1,800

5 Expected Costs

The proposed full research budget is **\$1,800**.

The budget has two distinct purposes:

1. **Immediate experimental requirement: \$300.** This is the near-term amount required to resume large-scale dataset generation and validation.
2. **Full 3-month research requirement: \$1,500.** This includes the AI/LLM allocation and the proposed researcher stipend.

The existing RI-RAG MVP, sample datasets, benchmark code, and experimental infrastructure are already available. The immediate research bottleneck is therefore the resources required to scale the experimental dataset and validation process.

6 Experimental Methodology

The research will compare two systems under equivalent experimental conditions.

6.1 Conventional RAG

A conventional architecture in which the vector database stores embeddings together with the associated retrieval payload.

6.2 RI-RAG

A hybrid architecture in which:

1. The vector database stores the information required for semantic retrieval and record identification.
2. Vector search identifies candidate record identifiers.
3. Corresponding records are retrieved from the relational database.
4. The relational database remains the authoritative source for textual payload and structured metadata.

6.3 Controlled Variables

Where technically possible, both systems will use:

- the same embedding model;
- the same query set;
- equivalent hardware;
- equivalent dataset records;
- equivalent retrieval parameters;
- controlled warm-up procedures;
- repeated benchmark runs; and
- identical evaluation criteria.

Database state, caching behavior, and initialization effects will be documented and controlled as far as practical.

6.4 Principal Measurements

The experiments will measure:

- Retrieval latency.
- Average latency.
- Median latency.
- P95/P99 latency where applicable.
- Peak RAM consumption.
- Baseline RAM consumption.
- Incremental RAM consumption attributable to the RAG system.
- Vector-storage footprint.
- Total database/storage footprint.
- Retrieval precision and recall.
- End-to-end retrieval correctness.
- Content–summary semantic similarity.
- Scaling behavior from 10k to 100k records.

Particular attention will be given to the distinction between vector-store memory reduction and total system RAM reduction.

A reduction in vector payload size does not automatically imply an equivalent reduction in overall RAM consumption because the database engine, operating system, indexes, caches, embedding model, and other components also consume memory.

This distinction is therefore a central research question.

7 Experimental Questions

7.1 RQ1 — Retrieval Performance

Can RI-RAG maintain retrieval latency that is competitive with conventional vector-based RAG as the dataset increases from 10,000 to 100,000 records?

7.2 RQ2 — Memory Consumption

Does reducing payload stored in the vector database produce a measurable reduction in overall RAM consumption of the complete RAG system?

7.3 RQ3 — Scalability

How do retrieval latency, memory consumption, and storage requirements change as the number of records increases?

7.4 RQ4 — Semantic Summary Quality

Do generated `semantic_summary` fields maintain a consistent semantic relationship with corresponding `content` fields at both dataset scales?

7.5 RQ5 — Architectural Trade-offs

What trade-offs exist between memory consumption, retrieval latency, storage requirements, and the additional relational-database retrieval step introduced by RI-RAG?

8 Project Timeline and Key Milestones

8.1 Milestone 0 — RI-RAG Prototype and Preliminary Evaluation [COMPLETED]

This milestone has been completed.

Activities included:

- Development of the RI-RAG MVP.
- Implementation of the relational-vector retrieval workflow.
- Construction of the initial controlled corpus.
- Conventional RAG baseline implementation.
- Initial vector-store comparison.
- Isolated-process memory evaluation.
- MySQL process monitoring.
- Retrieval accuracy and evidence-grounding evaluation.

Deliverable: Working RI-RAG MVP, initial benchmark infrastructure, preliminary experimental results, and research manuscript.

8.2 Milestone 1 — Efficiency Improvement and Experimental Infrastructure [COMPLETED]

This milestone has also been completed as part of the initial research phase.

The RI-RAG implementation and experimental infrastructure were developed and refined using the researcher’s existing computing resources.

Activities completed include:

- Profiling the RI-RAG implementation.
- Improving database access patterns.
- Implementing and validating vector retrieval.
- Implementing relational record hydration.
- Reducing unnecessary vector-layer payload storage.
- Establishing process-level memory measurement.
- Monitoring the MySQL process required by RI-RAG.
- Establishing the conventional RAG comparison baseline.
- Establishing a common evaluation protocol.
- Evaluating retrieval correctness and evidence grounding.

The completed benchmark infrastructure produced preliminary evidence including a 24.70% reduction in measured materialized Chroma representation and a 12.65% reduction in combined Python and MySQL peak RSS increase under the evaluated 13-record workload.

Deliverable: Completed RI-RAG implementation, comparative benchmark infrastructure, and preliminary experimental results.

8.3 Milestone 2 — 10,000-Record Dataset [PENDING FUNDING]

The next research phase will begin with generation and validation of approximately **10,000 records**.

The current research is paused at this stage pending approximately **\$300 in funding** for AI-assisted dataset generation, refinement, and validation.

Activities will include:

- AI-assisted dataset generation.
- Schema validation.
- Relational consistency validation.
- Content–summary semantic validation.

- Embedding generation.
- Database ingestion.
- Large-scale benchmark execution.
- Repeated memory and latency measurements.

Deliverable: Validated 10k-record benchmark dataset and comparative experimental results.

8.4 Milestone 3 — 100,000-Record Dataset

The experiment will subsequently scale to approximately **100,000 records**.

The same evaluation protocol will be applied to determine whether behavior observed at 10,000 records persists at substantially larger scale.

Deliverable: Validated 100k-record benchmark dataset and scalability results.

8.5 Milestone 4 — Comparative Analysis

The experimental results will be statistically analyzed to determine:

- Whether RI-RAG maintains competitive retrieval latency.
- Whether vector-storage reduction persists with increasing dataset size.
- Whether the reduction translates into lower total RAM consumption.
- How relational database hydration affects end-to-end latency.
- Whether content–summary semantic similarity remains consistent at scale.
- At what dataset scale architectural differences become measurable.
- What trade-offs exist between memory consumption and retrieval latency.

Deliverable: Statistical analysis of scalability, memory, storage, retrieval, and semantic-quality results.

8.6 Milestone 5 — Research Publication

The results will be consolidated into an expanded research paper documenting:

- RI-RAG architecture.
- Experimental methodology.
- Dataset-generation methodology.
- Content–summary validation.
- Experimental setup.
- Resource measurements.

- Scalability results.
- Limitations.
- Reproducibility considerations.

Deliverable: Expanded research manuscript based on the completed large-scale evaluation.

9 Anticipated Outcomes

The primary anticipated outcome is a rigorous evaluation of whether RI-RAG provides measurable resource and architectural differences at larger database scales.

The project will investigate whether RI-RAG can provide:

- Retrieval performance competitive with conventional vector-based RAG.
- Reduced vector-store memory requirements.
- Potential reductions in overall RAG system resource consumption.
- Improved traceability through an authoritative relational data layer.
- Explicit separation between semantic retrieval and authoritative data storage.
- Access to structured relational metadata during retrieval.
- A controlled methodology for evaluating automatically generated RAG datasets.

The research will **not assume that these benefits necessarily occur**. Instead, the experiments will determine their magnitude and identify the conditions under which they occur.

An important outcome will therefore also be a clear characterization of RI-RAG’s limitations and trade-offs.

If relational database hydration introduces latency that offsets measured memory benefits, this will be reported as an experimental finding. Similarly, if vector-store memory reduction does not translate into proportional total-RAM reduction, the research will quantify that difference.

10 Expected Scientific Contribution

The proposed research aims to contribute an experimentally validated architecture for separating semantic indexing from authoritative payload storage in RAG systems.

The contribution is particularly focused on the relationship between:

semantic retrieval → relational hydration → memory consumption → end-to-end retrieval performance.

The research will also provide a systematic approach for validating the relationship between source content and automatically generated semantic summaries.

Rather than evaluating only retrieval accuracy, the study will examine RAG as a systems problem involving:

- Memory.
- Storage.
- Retrieval latency.
- Semantic representation.
- Data traceability.
- Scalability.

The resulting benchmark will provide quantitative evidence regarding the scalability characteristics of RI-RAG and conventional RAG under controlled dataset sizes.

11 Publication Plans

The current RI-RAG MVP and its initial experimental results are planned for submission to **arXiv** as a preliminary research contribution.

Following completion of the larger-scale experiments, the expanded research will be developed into a substantially more comprehensive manuscript for potential submission to an appropriate **peer-reviewed venue, including Springer venues where the scope and experimental results are appropriate.**

The larger study will provide additional evidence including:

- 10,000-record evaluation.
- 100,000-record evaluation.
- Resource profiling.
- Memory analysis.
- Retrieval-latency analysis.
- Content–summary semantic validation.
- Scalability analysis.
- Statistical comparison between conventional RAG and RI-RAG.

12 Collaboration Opportunities

The project is **open to collaboration.**

Potential collaborators may contribute to:

- Dataset construction and validation.
- RAG benchmarking.
- Database and systems optimization.

- AI infrastructure.
- Statistical analysis.
- Reproducibility.
- AI systems research.
- Academic publication.

The project is particularly open to collaboration with researchers and engineers working on RAG systems, vector databases, relational databases, AI infrastructure, information retrieval, and efficient AI systems.

13 Budget Summary

Table 4: Project budget summary.

Item	Amount
AI assistance and LLM usage	\$300
3-month research stipend	\$1,500
Total	\$1,800

The immediate objective is to secure approximately **\$300** to resume the next experimental phase and scale the evaluation from the completed 13-record prototype to 10,000 and subsequently 100,000 records.

The broader **\$6,300** budget represents the proposed resource requirement for a sustained 12-month research program.

The central scientific objective is to establish, through controlled experiments, whether RI-RAG can maintain competitive retrieval performance while reducing resource consumption and providing the traceability benefits of an authoritative relational data layer.